



Do repair decisions matter?

Unobserved decisions and heterogeneity in a dynamic discrete choice model

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Objectives

- Propose a model of repair in a sintetized way
- Propose an identification strategy

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The data

The model



The data

First we begin to describe the data at hand so the model is put into more context. It consists three databases spanning 3 years. The databases are

- **Counts** This is the frequency of trade of automobiles
- **demo** The demography of the types of consumer
- **Counts scrap** The frequency of scrapped vehicles
- **car attributes** The attributes of the cars considered

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Counts database

This one is at the year (t), consumer type (τ), model (j) and age of the car (a) level. The observations describe what action was taken (choosing some car or no car) and the odometer from that car that they acquired (average).

tau	s_car_type	s_car_age	d_car_type	d_car_age	year	count	d_fuel_price	d_km
1	-1	-1	-1	-1	1996	193719		
1	-1	-1	1	0	1996	2072	8.064856309	37.5976122
1	-1	-1	1	1	1996	251	7.772069358	41.7049229
1	-1	-1	1	2	1996	470	7.548685246	45.1266583
1	-1	-1	1	3	1996	379	7.497405956	45.0238089
1	-1	-1	1	4	1996	616	7.3363444	46.1546132
1	-1	-1	1	5	1996	591	7.385188871	40.2615405
1	-1	-1	1	6	1996	500	7.35107329	40.3113139



Demo

tau	year	count	age	bigcity1	nkids	real_inc	single	wd
1	1996	558820	53.5479627	0.221735085	0.515192728	308282.229	0	0
1	1997	560874	53.7359951	0.220323994	0.513120594	306139.642	0	0
1	1998	556487	54.0997472	0.216666337	0.505760242	306719.856	0	0
1	1999	557468	54.2750041	0.215650046	0.502497004	305531.23	0	0
1	2000	570647	54.1293926	0.214975282	0.513036956	306472.441	0	0
1	2001	554658	54.3955293	0.215419231	0.506450822	307827.073	0	0
1	2002	501088	54.8688254	0.221513986	0.486028402	315943.919	0	0
1	2003	491485	55.3376654	0.220383125	0.475788681	315791.084	0	0



Scrap count

This includes the **probability of scrap** gotten from the frequency.

tau	s_car_type	s_car_age	year	count	pr_scrap
1	-1	-1	1996	246590	
1	-1	-1	1997	218613	
1	-1	-1	1998	187052	
1	-1	-1	1999	163060	
1	-1	-1	2000	158377	
1	-1	-1	2001	152714	
1	-1	-1	2002	144254	
1	-1	-1	2003	137420	
1	-1	-1	2004	134147	



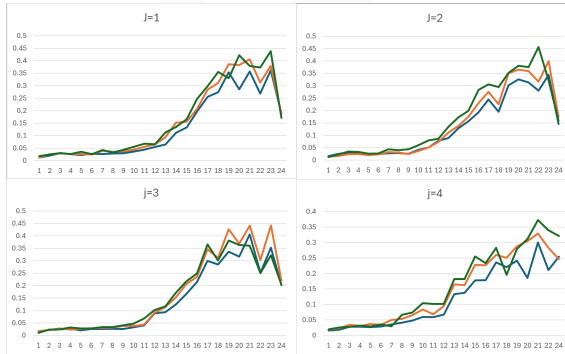
Car attributes

car_type	car_age	count	beta	diesel	fuel_eff	price_new	total_weight_tons	year
1	0	24865	0.875326844	0.001246732	13.7061679	154066.032	1.493399775	1996
1	1	21947	0.870945298	0.001230236	13.9165905	145674.992	1.457680574	1996
1	2	22987	0.871481819	0.000261017	13.5778051	147875.883	1.460712182	1996
1	3	13663	0.871965033	0.000439142	13.4106866	152257.492	1.474507624	1996
1	4	18206	0.872423094	0	13.2459056	152747.838	1.468139769	1996
1	5	16768	0.870438723	0	13.25067	147081.227	1.440703615	1996
1	6	14446	0.871168067	0	13.3807918	133781.246	1.398775871	1996
1	7	13329	0.871814306	7.50244E-05	13.2920363	127288.871	1.380044878	1996
1	8	15978	0.871467248	6.25861E-05	13.0407892	127001.514	1.378141873	1996
1	9	17591	0.871357712	0	12.7091069	128888.338	1.386442585	1996
1	10	21184	0.871033801	0	12.950262	115028.71	1.334518646	1996
1	11	22498	0.870663831	0	12.6521723	116437.58	1.338777269	1996
1	12	12532	0.869899996	7.97957E-05	12.3779782	117842.075	1.338200473	1996
1	13	10846	0.86982826	0	12.0311687	119894.203	1.348257791	1996
1	14	3581	0.869276245	0	11.8792061	118184.301	1.336133762	1996
1	15	4721	0.869616345	0	11.9623782	107598.601	1.303658123	1996
1	16	3197	0.869668644	0	11.9412058	100311.095	1.280860494	1996



Why now the data?

This model relies heavily in the variation and heterogeneity coming from scrap probability. We can observe that in the data. There are estational trends (it is mentioned in the paper). The x axis is probability of scrap and the x axis is age of car. The different lines are types of consumers.





The data

The model



The model: Introduction

- We live in a world with N consumers which have preferences $\succsim$ represented by a concave stationary utility function $u(\cdot)$.
- They live an infinite amount of periods
- They face decisions $B(z) \subseteq D$ that are available depending on the state $z \in Z$ and choose any $d \in B(z)$.
- The state space is defined by $z = (j, a, s)$, $j \in J$ is the car model, $a \in A$ is the age of the car and s the probability of scrap. This last one is a continuous variable.
- Consumers choose whether to buy, sell, keep and/or repair/give maintenance to their vehicles.
- Since utility and primitives are stationary, the distribution of ownership of cars q is stationary (time invariant).



The model: Consumers

- Consumers have a discount rate $0 < \beta < 1$
- They are represented by the following problem

$$\max_{d_0, d_1, \dots} U(d_0) + \sum_{m=1}^{\infty} \beta^m \mathbb{E}[U(d_m)]$$

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The model: Cars dynamics

- Cars can be from age 1 to age $\bar{a}$ which is denoted terminal age. They are also prone to accidents which sends them to the scrapyard with probability s , which sends them to terminal age.
- Cars increase deterministically in age every period by the probability $1 - s$
- Once a car ages to the terminal age $\bar{a}$, it gets sent to the scrapyard, therefore leaving the sample.
- Cars enter the sample by consumers buying new cars. Supply of new cars is perfectly elastic.



The model: Consumers

- The utility per car and decision is random, so it has an iid extreme value type I error. The error is decision specific.
- The types of consumers are differentiated by their wealth effect coefficient μ_τ .
- Inside the Wealth effects we find transaction costs, earnings from selling and losses from buying and repairing

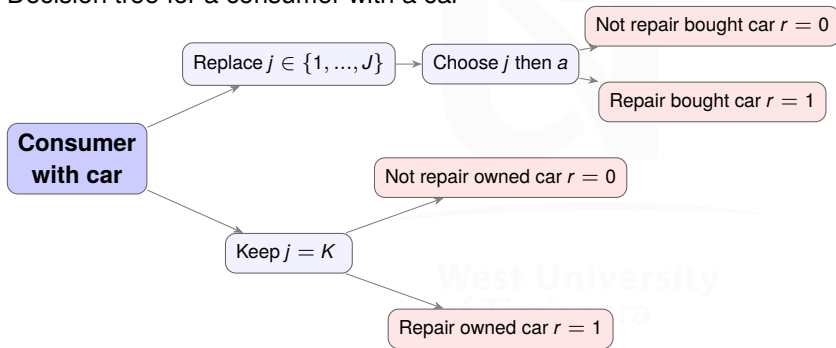
Utility for choice that got consumer with model j , age a , scrap prob s in a single period

$$U_{\tau,d} = u(j, a) - \mu_\tau(\text{Wealth effects}) + \varepsilon_d$$



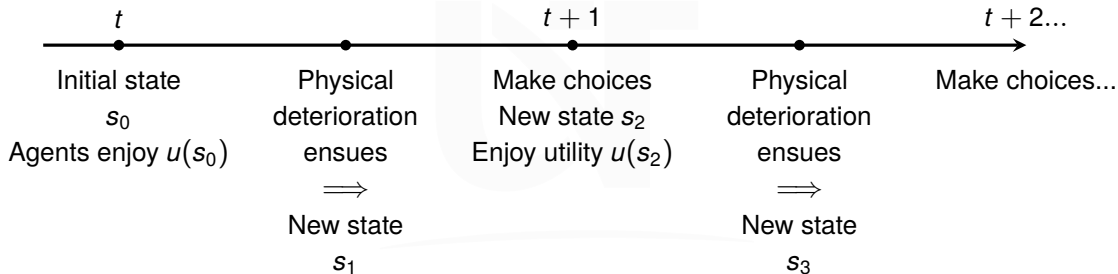
The model: Decision tree

Decision tree for a consumer with a car





The model: When are decisions made?





Decisions and state variables

- We have set up the primitives, but we haven't talked about how repair affects in any way the utility function.
- Consumers cannot scrap their cars through their own choice, instead they have to sell it or let it gain dust until it is sent to the scrapyard or some accident takes it out of commission. I argue that this is what most commonly happens in the real world.
- The effect that repair/maintenance has in this model is **lowering the probability of being sent to terminal age**
- It is important for the identification theorems for s to remain continuous.



Scrap and repair

- Let the state variable s_t evolve according to the following process

$$s_{t+1} = m(j, a, \underline{P}, k, r) + \eta, \quad \eta \sim N(0, \sigma_\eta)$$

Where $\underline{P}$ is the earnings from scrapping, k is the odometer and r is the repair decision.

- This follows Hu & Xin (2024) to identify unobserved decisions.
- We can try to parametrize it so it is easier for estimation



The model: Value function

We follow the traditional approach by defining the value function as

$$V(j, a, s) = \max_{d \in Z} [v(j, a, s, d) + \varepsilon(d)]$$

Where d is an element from the state space Z or the subset B available from the state j, a, s . We also included the choice specific value function $v(\cdot)$ that is defined as:

$$v(j, a, s, d) = E [U(d) + \beta V(d + 1)]$$



The model: Value function

- Overall, the setup is the same. I will keep it brief for the sake of simplicity
- The value function is

$$V(j, a, s) = \max \left[\begin{array}{l} v(j, a, s, k) + \varepsilon(k) \\ \max_{\substack{i \in \{1, \dots, J\}, \\ b \in \{0, 1, \dots, \bar{a}-1\}, \\ r \in \{0, 1\}}} [v(j, a, s, i, b, s', r) + \varepsilon(i, b, s, r)] \end{array} \right]$$

Keep

Replace and repair



Choice specific value function

Developing the previous function more, we explore how it looks with different decisions

- Being in state j, a, s and choosing to sell your car and buy and repair another car i, b

$$v(j, a, s, i, b, 1) = u(i, b) - \mu[P_{ib} - P_{ja} + T_b(i, b) + T_s(j, a) - P_r] + \beta[(1 - s(1))EV(i, b + 1, s') + s(1)EV(i, \bar{b}, 1)]$$

Where s' is next periods scrap probability without the effect of maintenance.

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Log-sum



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Probabilities

- We get the probabilities with the usual approach of multinomial logit as utility is random.



Equilibrium

- This model has prices in the secondary market as endogenous.
- We have to change the definition of demand and supply as we took options out.
- By having the trade transition matrix $\Omega(P)$ and the physical transition matrix Q we can find also the stationary distribution vector q that describes the economy.

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Equilibrium

- We define equilibrium as the price vector P^* and distribution vector q^* that satisfy

$$\begin{aligned}ED(P^*, q^*) &= 0 \\ q^* &= q^* R(P^*) \Omega(P^*) Q\end{aligned}$$



Identification!!

